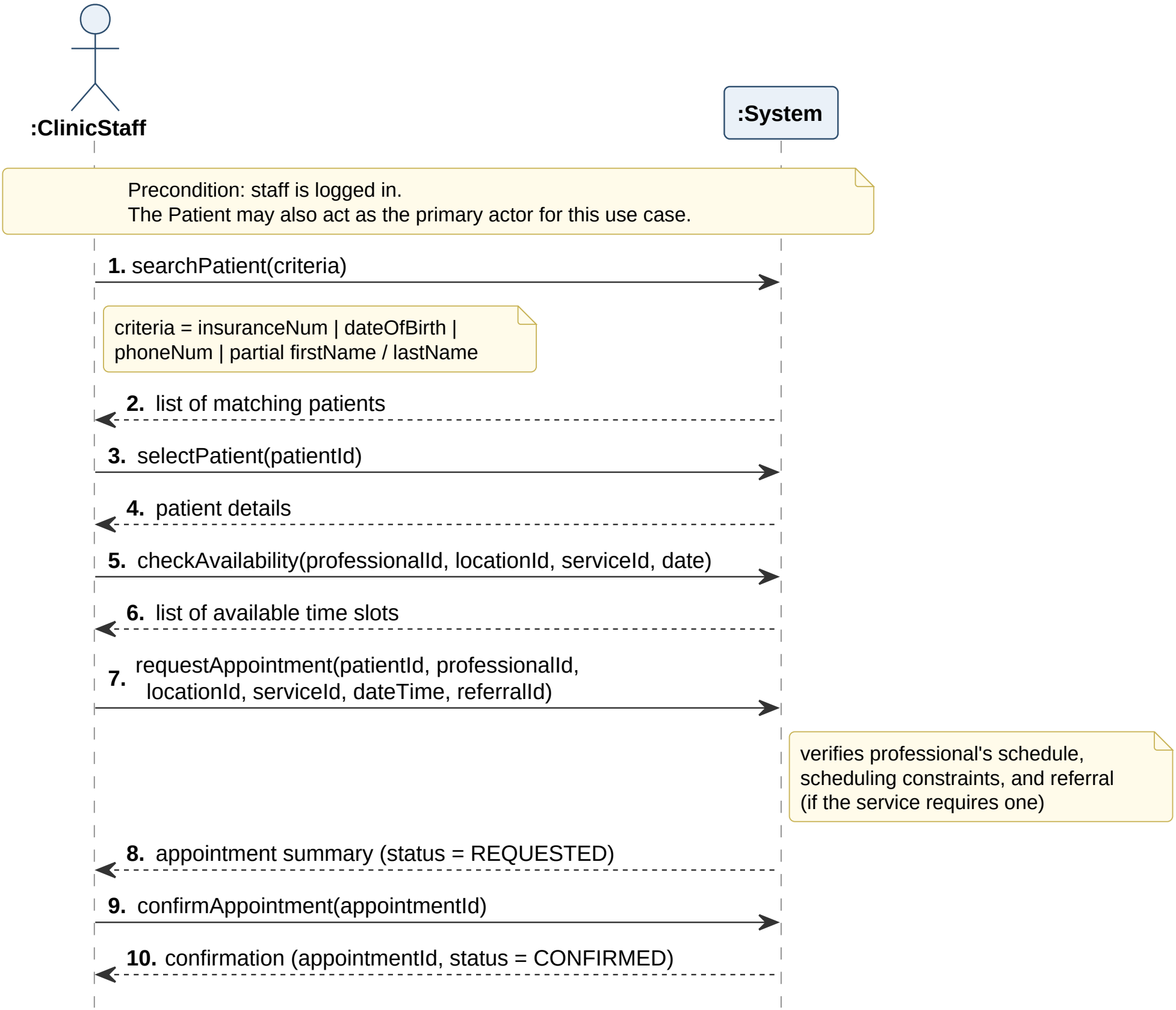
SOEN 342 – Iteration 1: System Sequence Diagrams

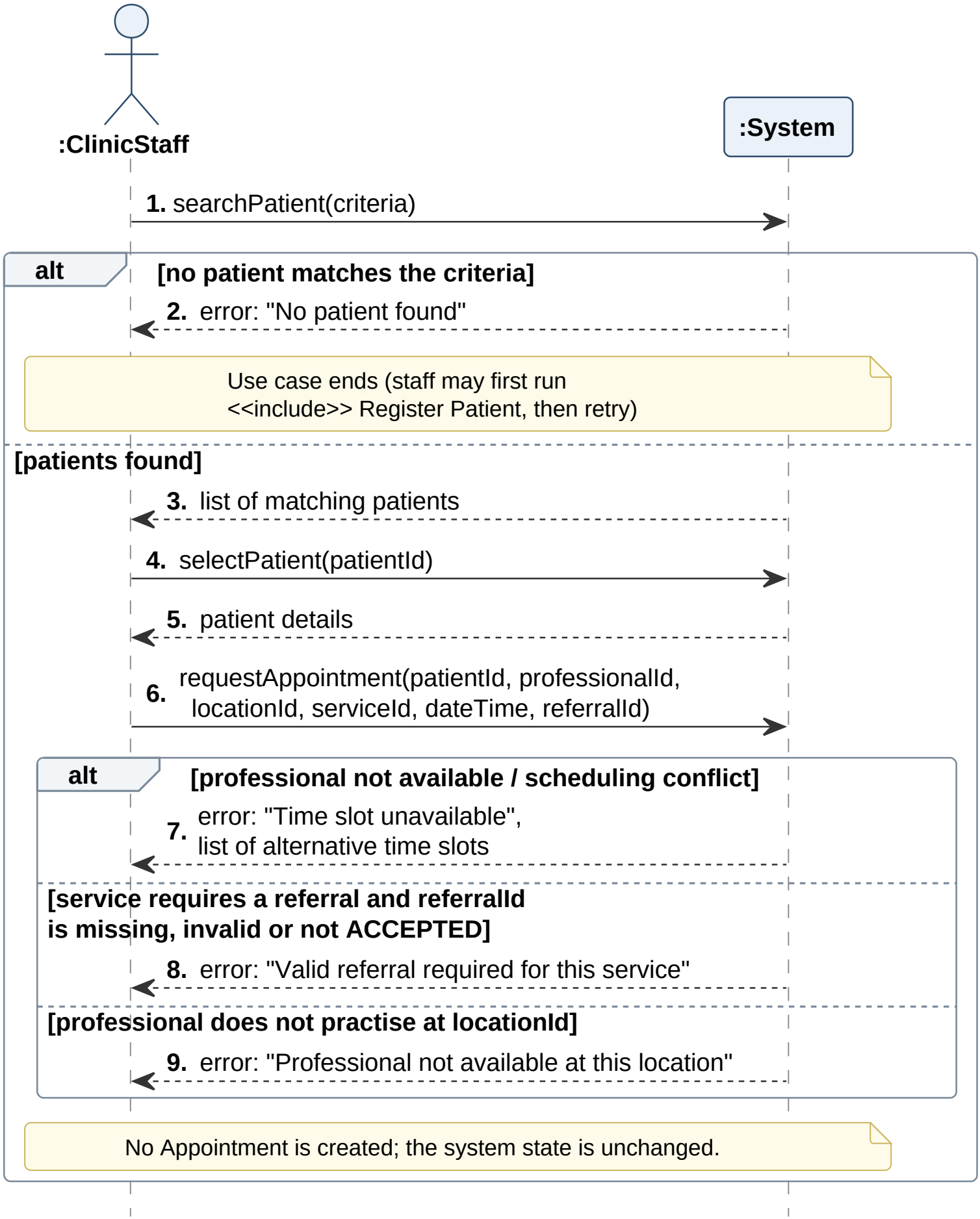
Each critical use case has a main success scenario SSD and a failure (alternative flow) SSD. The system is treated as a black box; arrows are system operations and dashed arrows are returned results.

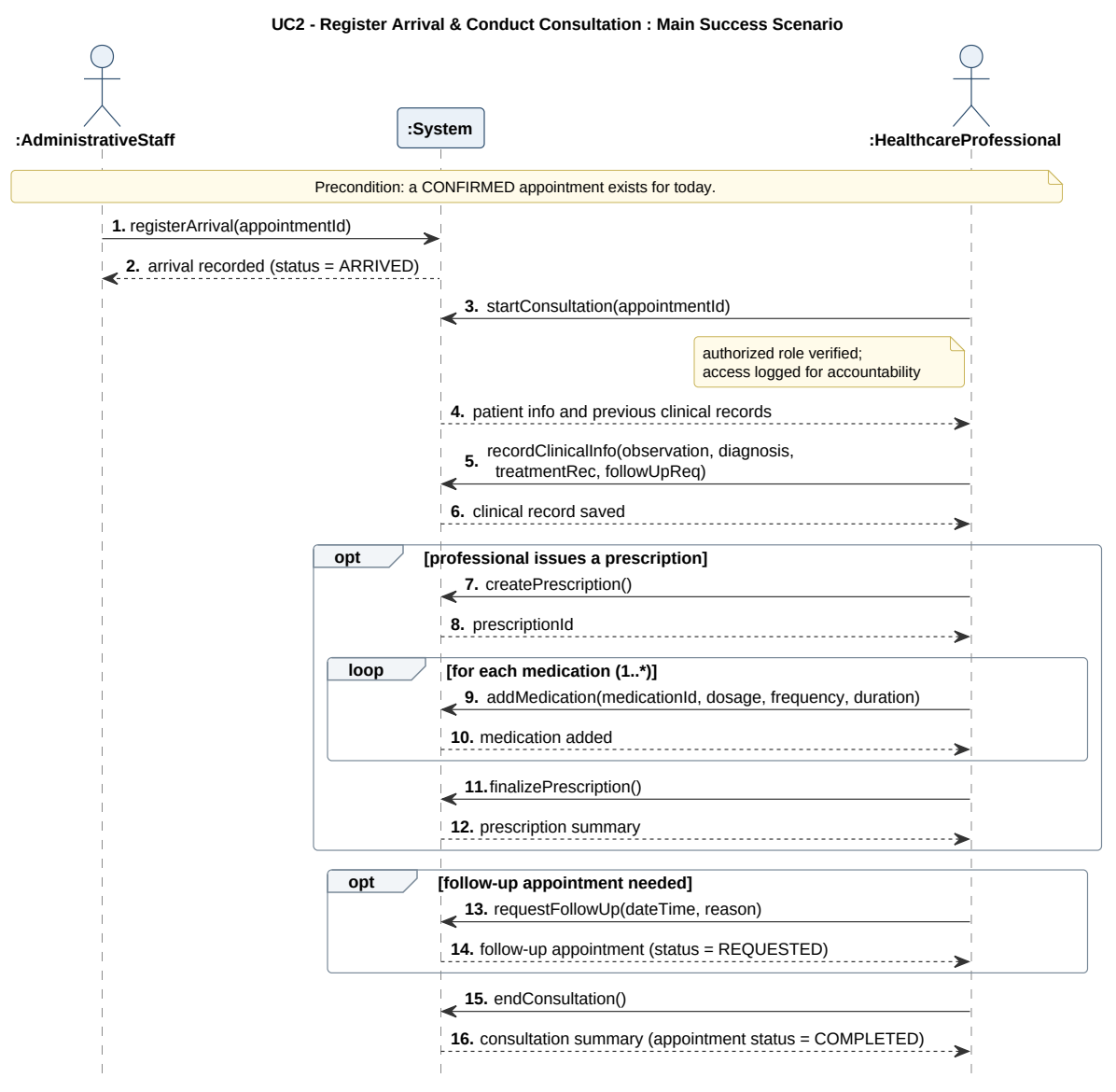
Diagram	File	Why it was made	Scenario shown
UC1 – Book Appointment (Success)	UC1_BookAppointment_Success	Booking is the core workflow of the clinic and the entry point for every visit; it must enforce schedules, locations and referral rules.	Staff searches a patient (insurance #, DOB, phone, partial name), checks a professional's availability, requests a time slot and confirms it. Appointment ends CONFIRMED.
UC1 – Book Appointment (Failure)	UC1_BookAppointment_Failure	Shows how the system protects scheduling integrity and responds when a booking cannot be made.	Patient not found (register first); time slot unavailable (alternatives returned); missing / invalid referral; professional not practising at the chosen location. No appointment is created.
UC2 – Arrival & Consultation (Success)	UC2_ConductConsultation_Success	Captures the clinical heart of the system: recording care, prescriptions and follow-ups, involving two actors.	Admin registers arrival; professional starts consultation, records observation / diagnosis / treatment / follow-up, optionally issues a prescription (1..* medications) and requests a follow-up; consultation completed.
UC2 – Arrival & Consultation (Failure)	UC2_ConductConsultation_Failure	Demonstrates the privacy / role-based access requirement and validation of clinical data.	Invalid or cancelled appointment at arrival; admin staff denied access to clinical records (logged); incomplete clinical record; medication not in catalog.
UC3 – Issue Referral (Success)	UC3_IssueReferral_Success	Referrals connect family physicians to specialists and are required before booking certain services.	Physician selects patient, creates a referral, enters requested specialty / professional, reason and priority, then submits it (status PENDING) for later booking.
UC3 – Issue Referral (Failure)	UC3_IssueReferral_Failure	Shows that incomplete or invalid consultation requests are rejected.	Requested specialty / professional not offered; reason missing or invalid priority; submission blocked and referral stays DRAFT.
UC4 – Invoice & Payment (Success)	UC4_InvoicePayment_Success	Covers the financial requirement: uninsured services are billed to the patient.	Admin generates an invoice for a completed uninsured service and registers a payment; receipt returned, invoice PAID when balance is 0.
UC4 – Invoice & Payment (Failure)	UC4_InvoicePayment_Failure	Distinguishes RAMQ-insured vs uninsured services and validates payments.	Insured (RAMQ) service: no invoice; payment amount invalid; invoice already settled.

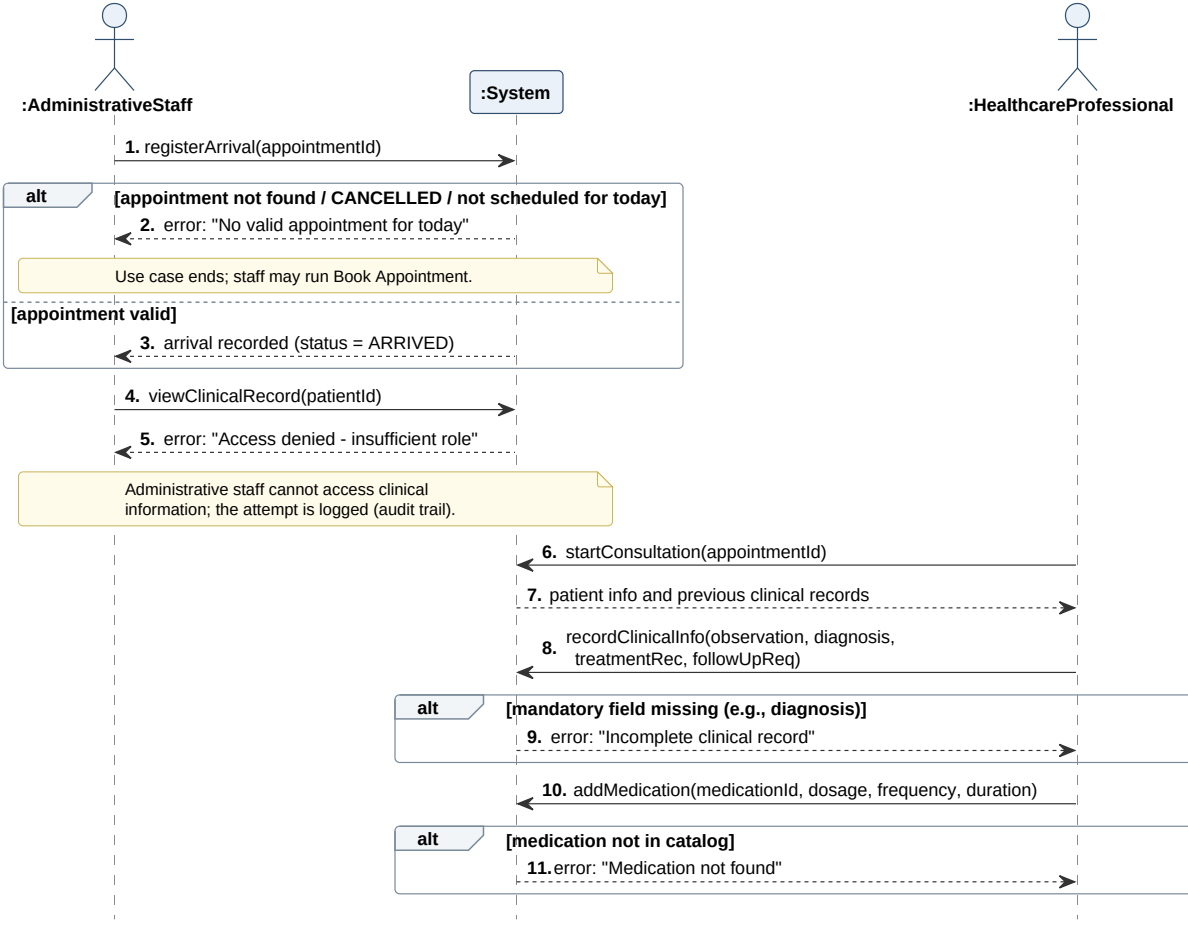
UC1 - Book Appointment : Main Success Scenario



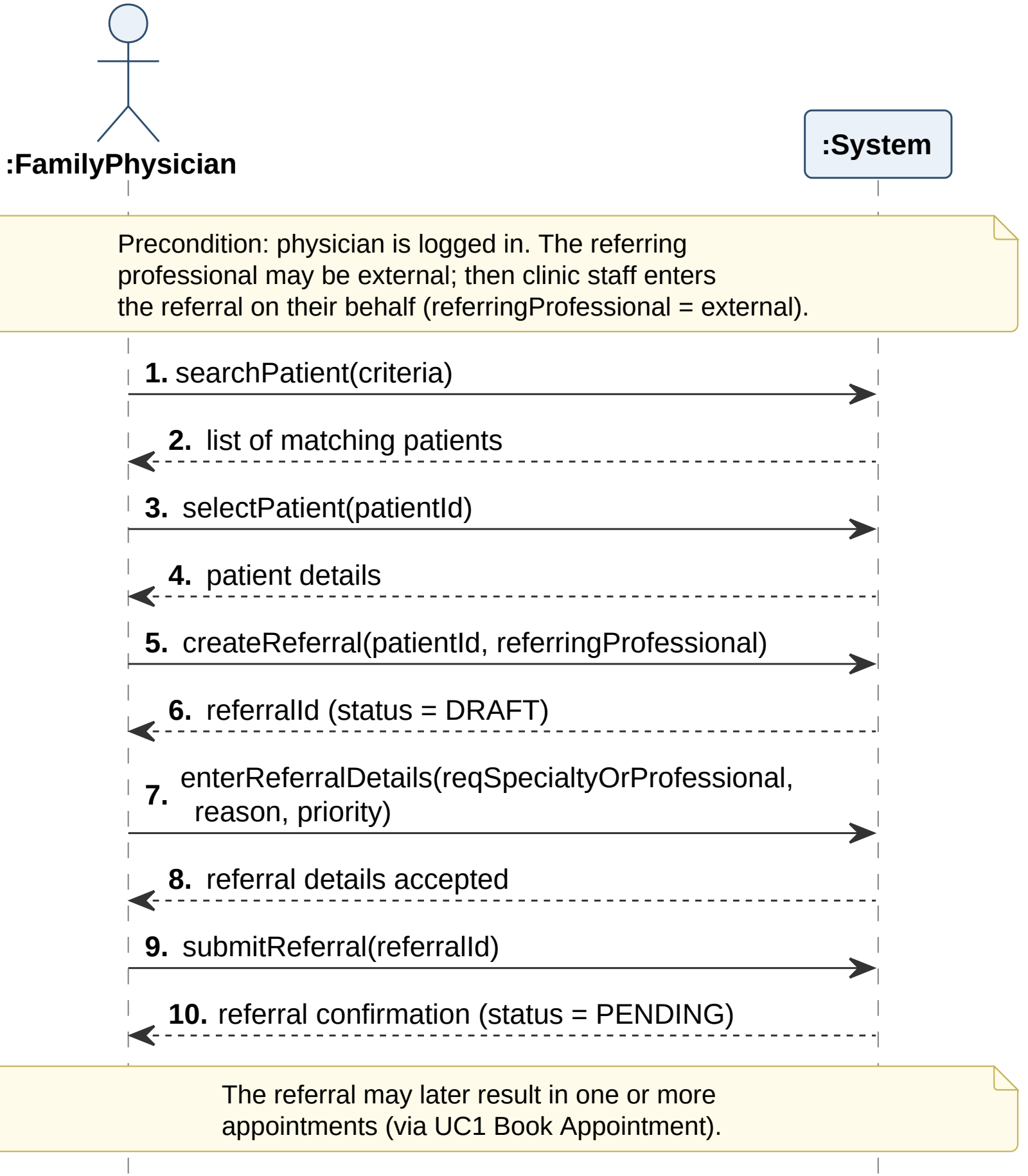
UC1 - Book Appointment : Failure Scenarios



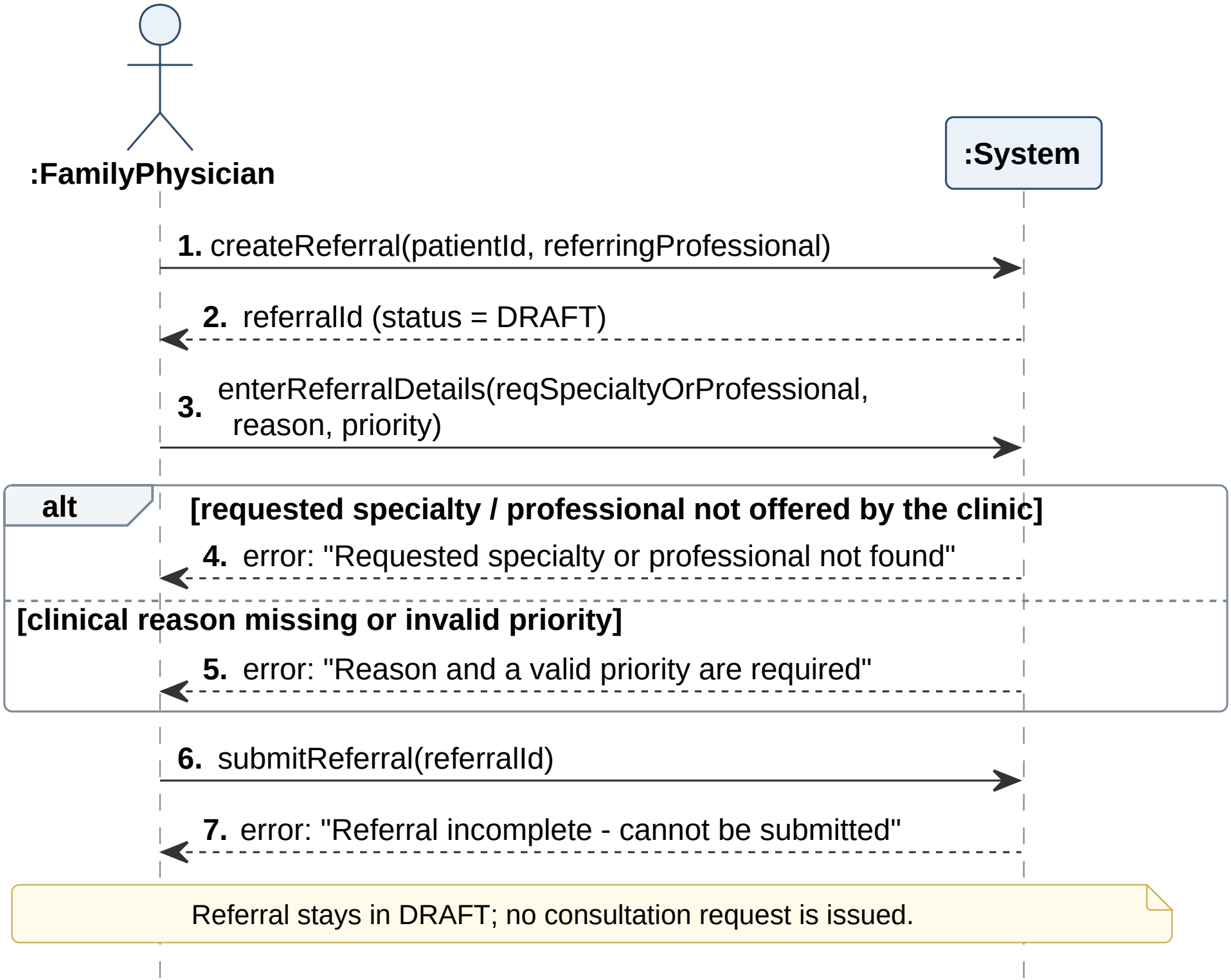

UC2 - Register Arrival & Conduct Consultation : Main Success Scenario
:AdministrativeStaff
:System
:HealthcareProfessional
Precondition: a CONFIRMED appointment exists for today.
1. registerArrival(appointmentId)
2. arrival recorded (status = ARRIVED)
3. startConsultation(appointmentId)
authorized role verified;
access logged for accountability
4. patient info and previous clinical records
5. recordClinicalInfo(observation, diagnosis, treatmentRec, followUpReq)
6. clinical record saved
opt
[professional issues a prescription]
7. createPrescription()
8. prescriptionId
loop
[for each medication (1..*)]
9. addMedication(medicationId, dosage, frequency, duration)
10. medication added
11. finalizePrescription()
12. prescription summary
opt
[follow-up appointment needed]
13. requestFollowUp(dateTime, reason)
14. follow-up appointment (status = REQUESTED)
15. endConsultation()
16. consultation summary (appointment status = COMPLETED)


UC2 - Register Arrival & Conduct Consultation : Failure Scenarios
:AdministrativeStaff
:System
:HealthcareProfessional
1. registerArrival(appointmentId)
alt
[appointment not found / CANCELLED / not scheduled for today]
2. error: "No valid appointment for today"
Use case ends; staff may run Book Appointment.
[appointment valid]
3. arrival recorded (status = ARRIVED)
4. viewClinicalRecord(patientId)
5. error: "Access denied - insufficient role"
Administrative staff cannot access clinical information; the attempt is logged (audit trail).
6. startConsultation(appointmentId)
7. patient info and previous clinical records
8. recordClinicalInfo(observation, diagnosis, treatmentRec, followUpReq)
alt
[mandatory field missing (e.g., diagnosis)]
9. error: "Incomplete clinical record"
10. addMedication(medicationId, dosage, frequency, duration)
alt
[medication not in catalog]
11. error: "Medication not found"

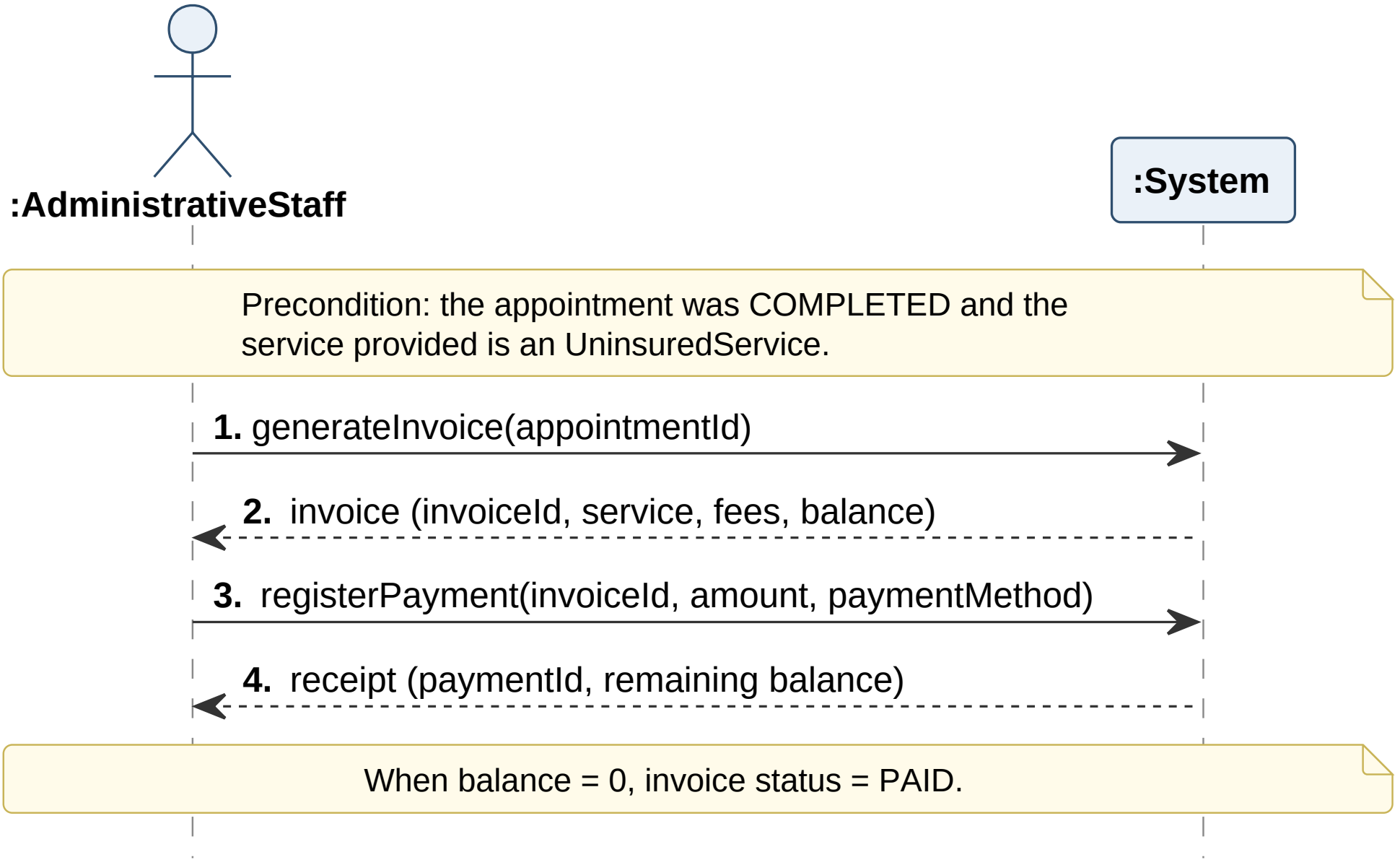
UC3 - Issue Referral (Consultation Request) : Main Success Scenario

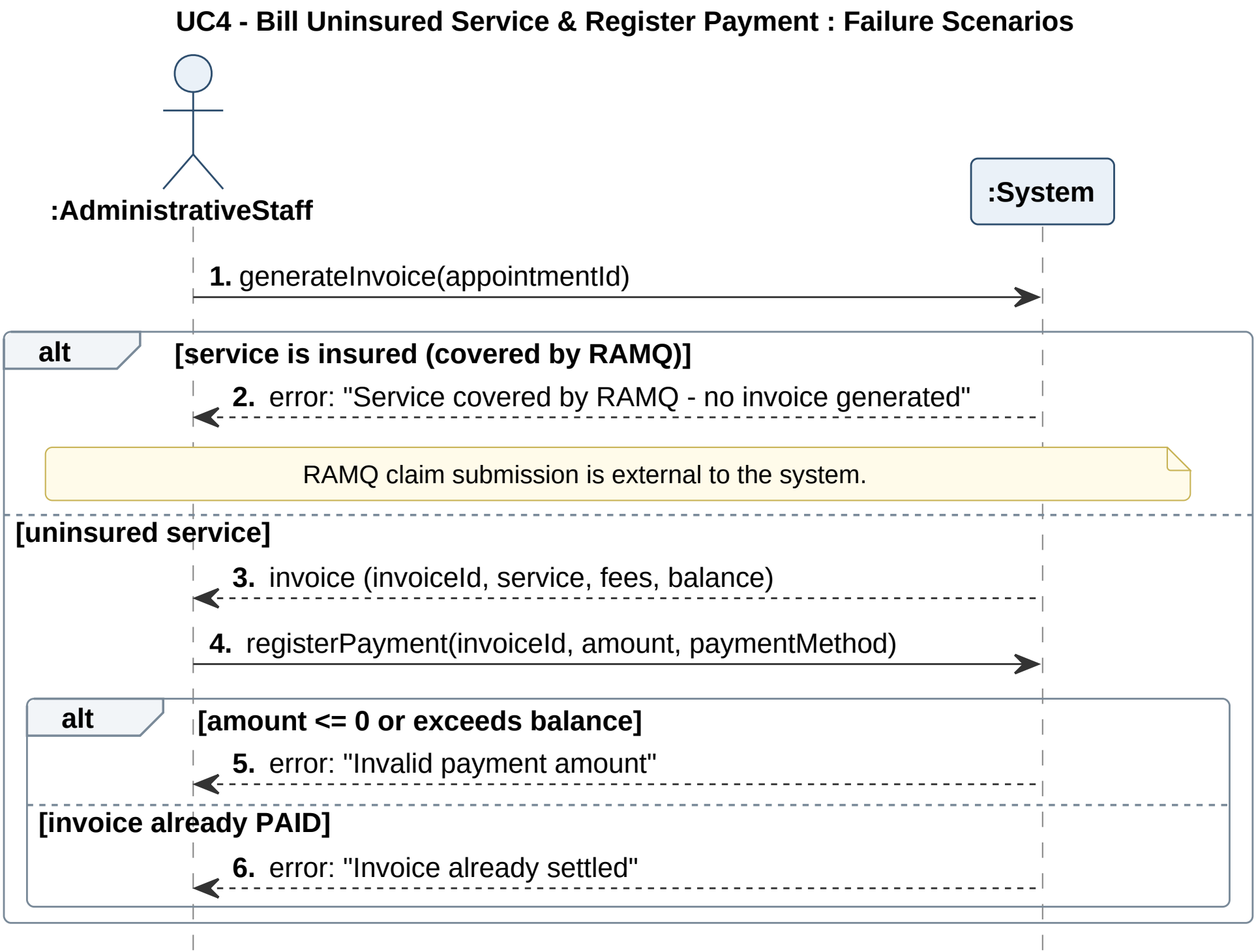


UC3 - Issue Referral (Consultation Request) : Failure Scenarios



UC4 - Bill Uninsured Service & Register Payment : Main Success Scenario




UC4 - Bill Uninsured Service & Register Payment : Failure Scenarios
:AdministrativeStaff
:System
1. generateInvoice(appointmentId)
alt
[service is insured (covered by RAMQ)]
2. error: "Service covered by RAMQ - no invoice generated"
RAMQ claim submission is external to the system.
[uninsured service]
3. invoice (invoiceId, service, fees, balance)
4. registerPayment(invoiceId, amount, paymentMethod)
alt
[amount <= 0 or exceeds balance]
5. error: "Invalid payment amount"
[invoice already PAID]
6. error: "Invoice already settled"